ROBOTICĂ - TEMA 2

FALSTAD CIRCUITS.

1. Series Circuit (5V DC)

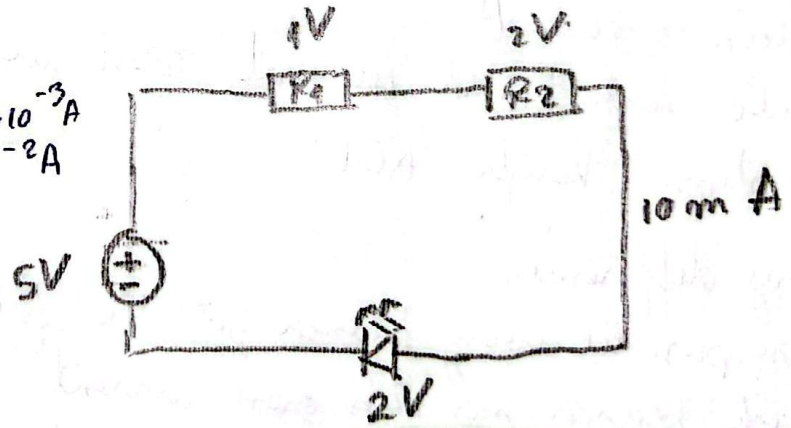
- Connect at least 2 resistors and 1 LED in series.
- Choose resistor values to obtain a current of 5-20 mA through the LED
- Calculate the current, voltage drops, and power dissipated by each resistor. Verify KVL

Formulas:

KVL: $V_{source} = V_{LED} + V_{R1} + V_{R2}$.

Ohm's law: $V_R = i \cdot R$

Power Dissipation: $P = i \cdot V_R = i^2 \cdot R$.



Target $i = 10\,mA = 10 \cdot 10^{-3} A = 1 \cdot 10^{-2} A$

$V_S = V_L + V_{R1} + V_{R2}$

$V_{R1} + V_{R2} = V_S - V_2$

$V_{R1} + V_{R2} = 5 - 2 = 3\,V$

So let's say $V_{R1} = 1\,V$, $V_{R2} = 2\,V$

$$i = \frac{V_S - V_{LED}}{R_1 + R_2} = \frac{5-2}{300} = 10\,mA$$

Ohm: $R = \frac{V}{i} \Rightarrow R_1 = \frac{1\,V}{1 \cdot 10^{-2} A} = 10^2\,\Omega = 100\,\Omega$

$R_1 + R_2 = 300\,\Omega$

$R_2 = \frac{2\,V}{1 \cdot 10^{-2} A} = 2 \cdot 10^2\,\Omega = 200\,\Omega$.

$\frac{3\,V}{1 \cdot 10^{-2} A} = 3 \cdot 10^2\,\Omega = 300\,\Omega$ ✓

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$P = V \cdot i = i^2 \cdot R$.

$P_{R1} = V_{R1} \cdot i = 1V \cdot 10\,mA = 1 V \cdot 10^{-2} A = 0,01\,W$

$P_{R2} = V_{R2} \cdot i = 2\,V \cdot 10\,mA = 2\,V \cdot 10^{-2} A = 0,02\,W$

$P_L = V_L \cdot i = 2\,V \cdot 10\,mA = 0,02\,W$.

KVL check: $V_{R1} + V_{R2} + V_L = -1 + 2 + 2 = 5\,V$

Obs: The bigger resistor heats up more because $P = i^2 \cdot R$ and while i is the same R differs, so double the resistance means double the heat

$P_S = V_S \cdot i = 5\,V \cdot 10\,mA = 5 \cdot 10^{-2} = 0,05\,W$.

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2 Parallel Circuit (5V)

- 2 parallel branches, each containing LED + resistor
- resistor values so the 2 LEDs carry different currents, each between 5-20 mA
- calc. each branch current, total current, resistor and power dissipation. Verify KCL

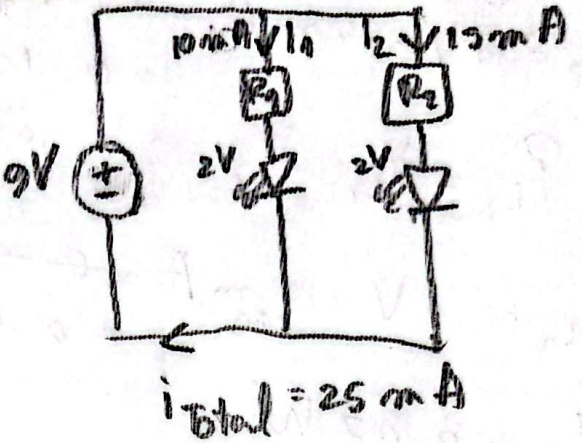
Theory differences:

- in parallel every branch gets the full source voltage (5V)
- each branch has it's own current
- KCL: $i_{Total} = i_1 + i_2$
- KVL still applies inside each branch: $V_S = V_R + V_{Led}$

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I will work with 2 red leds.



KVL for each branch ⇒

$V_S = V_{R_1} + V_{L_1} = V_{R_2} + V_{L_2}$

$9 = V_{R_1} + 2 \Rightarrow$

$\Rightarrow V_{R1} = V_{R2} = 9 - 2 = 7\ V$

Now Ohm: $R_1 = \frac{V_{R1}}{i_1} = \frac{7\ V}{10\ mA} = \frac{7\ V}{10 \cdot 10^{-3} A} = 7 \cdot 10^2\ \Omega = 700\ \Omega$

$R_2 = \frac{V_{R_2}}{I_2} = \frac{7\ V}{15\ mA} = \frac{7\ V}{15 \cdot 10^{-3} A} = \frac{7000}{15} \simeq 466,7\ \Omega$

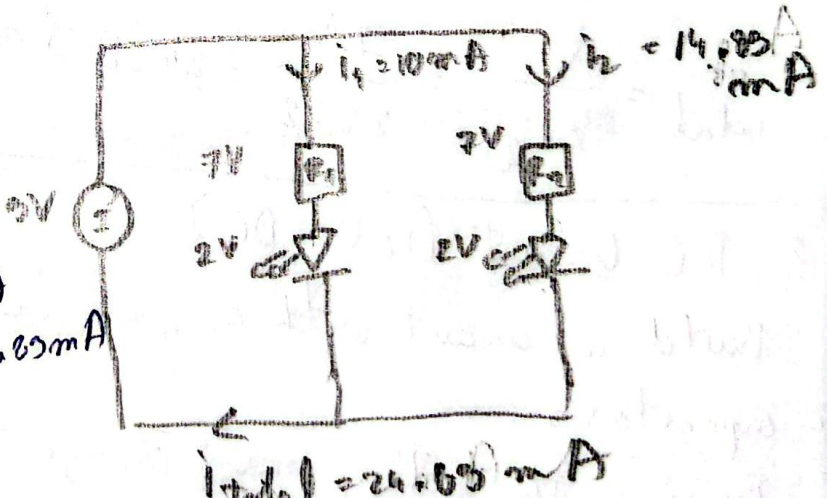
For the fun of it I will work with $R_2 = 470\ \Omega$

Branch currents

$I_1 = \frac{V_S - V_{L1}}{R_1} = \frac{9-2}{700} = \frac{7}{700} = 10\ mA$

$i_2 = \frac{V_S - V_{L_2}}{R_2} = \frac{9-2}{470} = \frac{7}{470} = \cancel{0,014 = 14mA}$

$0.01489 = 14.89 mA$



KCL:

$i_{total} = i_1 + i_2 = \cancel{10+14 = 24\ mA}$

$10 + 14.89 = 24.89\ mA$

KVL / branch:

$V_{R1} + V_{L1} = 7 + 2 = 9V = V_S$

$V_{R_2} + V_{L_2} = 7 + 2 = 9\ V = V_S$

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Power: $P = V \cdot i = V \cdot \frac{V}{R} = \frac{V^2}{R}$

$P_{R_1} = V_{R_1} \cdot i_1 = 7V \cdot 10\,mA = 7V \cdot 10^{-2}A = 0.07\,W$

$P_{R_2} = V_{R_2} \cdot I_2 = 7V \cdot 14\,mA = 7V \cdot 14,29 \cdot 10^{-3}A = 0.1013\,W$

$P_{L_1} = V_{L_1} \cdot i_{L_1} = 2V \cdot 10\,mA = 0.02\,W$

$P_{L_2} = V_{L_2} \cdot i_{L_2} = 2V \cdot 14,49\,mA = 0.02898\,W$

$P_{R_1} = \frac{7^2}{700} = \frac{7}{700} = 0.07\,W$

Obs: in parallel the voltage is the same / branch. so because $P = V^2/R$ the smaller resistor will draw more current and dissipate more power.

$R_{eq} = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{700 \cdot 470}{700 + 470} = \frac{329000}{1170} \simeq 281.2\,\Omega$

$I_{total} = \frac{V}{R_{eq}} = \frac{7V}{281.2} = 24,89\,mA$ ✓ (matches KCL)

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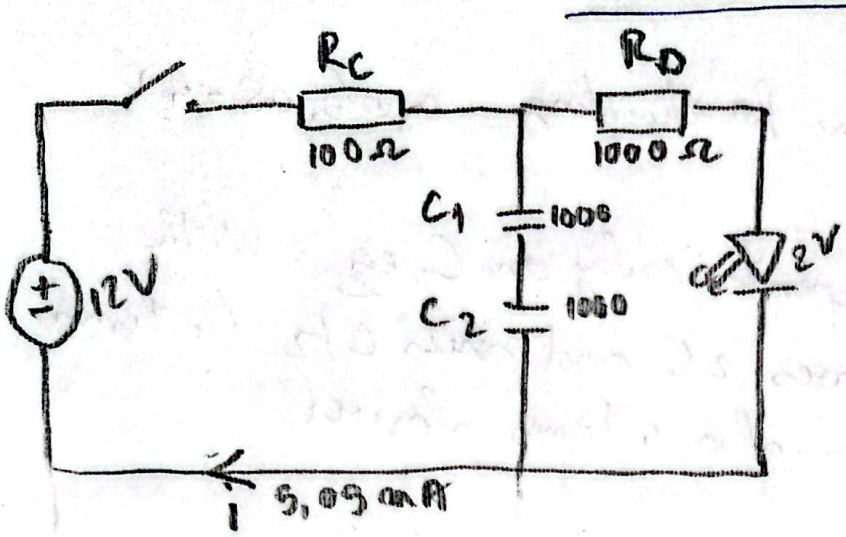
3. RC Circuit (12V DC)

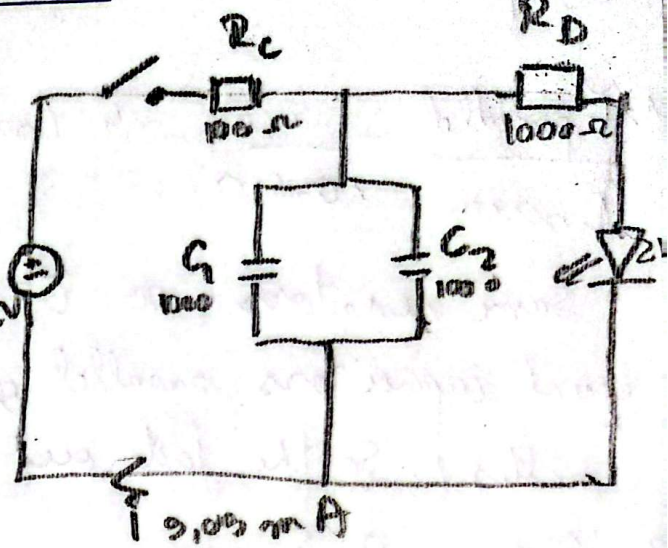
- Build a circuit with a push button, an LED, resistors and 2 capacitors.
- Pressing the button must charge the capacitors, releasing it must cause the LED to fade out gradually.
- Compare the same circuit with capacitors connected in series and in parallel
- Calc. the equivalent capacitance and RC for both.
- Choosing a capacitor voltage threshold and calc the time t needed to reach it during charging and discharging. State initial and target voltages & check time in Falstad.

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FALSTAD CIRCUITS





Common values:

$V_s = 12\,V$; $R_C = 100\,\Omega$ (charging resistor); $R_d = 1000\,\Omega$ (discharging resistor, protects LED); $C_1 = C_2 = 1000\,\mu F = 1 \cdot 10^{-3}\,F$

$V_{Led} = 2V$ (although red is 1.8 I choose it for ease of calc)

V_{max} (same for both, C doesn't appear)

$$I = \frac{V_s - V_{Led}}{R_C + R_d} = \frac{12-2}{100+1000} = \frac{10}{1100} = \frac{1}{110} = 0.00909\,A = 9,09\,mA$$

$$V_{max} = V_s - I \cdot R_C = 12 - 0,00909 \cdot 100 = 12 - 0,909 \simeq 11,09\,V$$

Series

$$C_{eq} = \frac{C_1 \cdot C_2}{C_1 + C_2} = \frac{1000 \cdot 1000}{1000 + 1000} = 500\,\mu F = 500 \cdot 10^{-6}\,F$$

$$\tau_C = R_C \cdot C_{eq} = 100\,\Omega \cdot 500 \cdot 10^{-6}\,F = 5 \cdot 10^{-2}\,\Omega = 0,05\,\Omega$$

$$\tau_D = R_D \cdot C_{eq} = 1000\,\Omega \cdot 500 \cdot 10^{-6}\,F = 5 \cdot 10^{-1}\,\Omega = 0,5\,\Omega$$

Full charge $= 5 \cdot \tau_C = 5 \cdot 0,05\,\Omega = 0,25\,\Omega$

Parallel

$$C_{eq} = C_1 + C_2 = 2000\,\mu F = 2000 \cdot 10^{-6}\,F$$

$$\tau_C = R_C \cdot C_{eq} = 100\,\Omega \cdot 2000 \cdot 10^{-6}\,F = 2 \cdot 10^{-1}\,\Omega = 0,2\,\Omega$$

$$\tau_D = R_D \cdot C_{eq} = 1000\,\Omega \cdot 2000 \cdot 10^{-6}\,F = 2\,\Omega$$

Full charge $= 5 \cdot \tau_C = 0,2\,\Omega = 1\,\Omega$

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Comparasion

1. $\frac{\tau_{parallel}}{\tau_{series}} = \frac{2\Omega}{0,5\Omega} = 4$ (same for charging $0,2/0,05 = 4$)

Same resistors, so τ depends only on C_{eq}. With equal capacitors, parallel gives $2C$ and series $C/2$ (4 times smaller). So the fade out is also 4 times slower

2. $\frac{\tau_d}{\tau_c} = \frac{0,5}{0,05} = 10$

Same C_{eq} so τ depends on R. $\frac{R_d}{R_c} = \frac{1000}{100} = 10$.

Capacitors charge 10 times faster than they discharge

3. 87% $\rightarrow 2 \cdot \tau_c = 0.1\Omega$ not full charge, short tap

$\boxed{\tau = R \cdot C}$

Threshold time: $V_{th} = 6V$ (around middle) $> 2V$ (led)

Charging: initial $0V$ $\rightarrow$ target $9V$

Discharge: initial $11V$ —//—

Formulas:

$$V(t) = V_{max}\left(1 - e^{\frac{-t}{\tau_c}}\right) \Big| \Rightarrow \frac{V_{th}}{V_{max}} = 1 - e^{-t/\tau_c} \Rightarrow$$

$V(t) = V_{th}$

$$\Rightarrow e^{-t/\tau_c} = 1 - \frac{V_{th}}{V_{max}} \quad / \ln.$$

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$$-\frac{t}{\tau_c} = \ln\left(1 - \frac{V_{th}}{V_{max}}\right)$$

$$t_{charge} = -\tau_c \cdot \ln\left(1 - \frac{V_{th}}{V_{max}}\right)$$

Common numbers:

$$\frac{V_{th}}{V_{max}} = \frac{6}{11.09} = 0.5410$$

$1 - 0,5410 = 0,4590$; $\ln(0,4590) \approx -0,7788$

Series ($\tau_c = 0.05\,s$):

$t = -0,05 \cdot (-0,7788) = 0,0389\,s \approx 38.9\,ms$

Parallel ($\tau_c = 0.2\,s$)

$t = -0,2 \cdot (-0,7788) = 0,1558\,s \approx 155.8\,ms$

Discharging formula:

- during discharge KVL around the loop (capacitor → R_D → LED) gives: $V_C = V_{Rd} + V_{Led} \Rightarrow V_{Rd} = V_C - V_{Led}$
- only the part above the led decays exponentially

$$V(t) - V_{Led} = (V_{max} - V_{Led})\, e^{-t/\tau_d}$$

$$V(t) = V_{th} : \frac{V_{th} - V_{Led}}{V_{max} - V_{Led}} = e^{-t/\tau_d} \quad /(\ln)$$

$$-\frac{t}{\tau_d} = \ln\left(\frac{V_{th} - V_{Led}}{V_{max} - V_{Led}}\right) \Rightarrow$$

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$$t_{discharge} = \tau_d \cdot \ln\left(\frac{V_{max} - V_{Led}}{V_{th} - V_{Led}}\right)$$

$$\frac{V_{max} - V_{Led}}{V_{th} - V_{Led}} = \frac{11.09 - 2}{6 - 2} = \frac{9.09}{4} = 2.2725$$

$\ln(2.2725) = 0.8209$

Series ($\tau_d = 0.5\Omega$): $t = 0.5 \cdot 0.8209 = 0.4104\Omega \approx 0,41\Omega$

Parallel ($\tau_d = 2\Omega$): $t = 2 \cdot 0,8209 = 1,6418\Omega \approx 1,64\Omega$

	Series	Parallel
Charging 0V -> 6V	38.9 mΩ	155.8 mΩ
Discharging 11.09V -> 6V	0.41Ω	1,64Ω

Parallel times are 4x the series times because C_{eq} is 4x bigger

Falstad Sim.

0V -> 6V.	42,30 ms	162 mΩ
11.09 -> 6V.	0.411Ω	1,78Ω

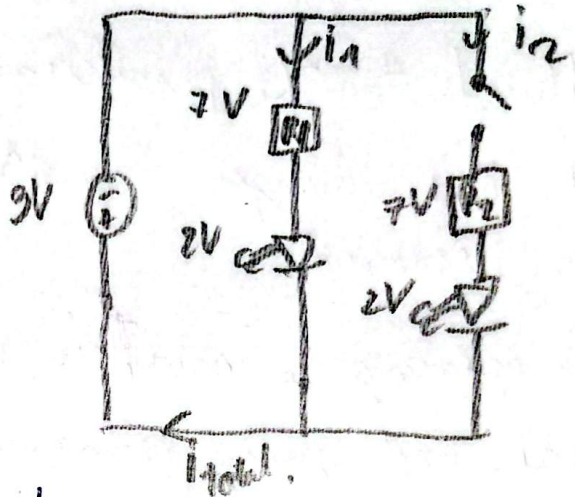
2,64
1,086

871 -
460
411

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Bonus.

Switch in branch 2 of the parallel circuit so branch 2 can be turned on and off



- Switch closed => same circuit as in ex 2.

$I_1 = 10\,mA$, $i_2 = 14.89\,mA$, $I_{total} = 24.89\,mA$

$R_{eq} = 281.2\,\Omega$

- Switch open => branch 2 is not a closed loop =>

=> $i_2 = 0\,mA$ => LED 2 is off.

Branch 1 is still connected

$V_S = V_{R_1} + V_{L_1}$

$9 = V_{R_1} + 2 \Rightarrow V_{R_1} = 7\,V$ (uncharged).

$i_1 = \frac{V_{R_1}}{R_1} = \frac{7\,V}{700\,\Omega} = 10\,mA$ (uncharged)

Since only R_1 is left => $R_{eq} = 700\,\Omega$ (it was $281.2\,\Omega$)

Check:

$I_{total} = \frac{7\,V}{700\,\Omega} = 10\,mA$ ✓

$P_S = V_S \cdot i_{total} = 9\,V \cdot 10\,mA = 90\,mW$ (was 224 mW)

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Conclusion.

Even if we open the switch the same current flows through branch 1 thanks to the parallelism prop

Only the KCL changes (24.89 -> 10 mA) because one path was removed.

In series opening a switch would stop the current everywhere, and every component would turn off.

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